



Current Final Term Questions

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Subjective

**1. How many daughter cells produce at the end of meiosis?
And what is the chromosome number in meiosis?**

At the end of meiosis, four daughter cells are produced. Each of these cells has half the number of chromosomes as the original cell, which means they are haploid. For humans, this means each daughter cell has 23 chromosomes.

2. Which flower is used for studying the phenomenon of cyclosis and why?

The flower used for studying the phenomenon of cyclosis is the water plant *Elodea*. This plant is used because its cells are large, transparent, and show clear cytoplasmic streaming, making it easy to observe the movement of the cytoplasm and organelles.

3. Types of bacteria according to the composition of the cell wall:

Bacteria can be classified based on their cell wall composition into:

Gram-positive bacteria: Have a thick peptidoglycan layer.

Gram-negative bacteria: Have a thin peptidoglycan layer and an outer membrane.

Acid-fast bacteria: Have a waxy, lipid-rich cell wall that does not readily take up Gram stain.

4. Stages of mitosis:

The stages of mitosis are:

- Prophase
- Metaphase
- Anaphase
- Telophase

5. Name of microscope used in labs:

The most common microscope used in labs is the compound light microscope.

6. Difference between mitosis and meiosis:

- Mitosis:

- Produces two genetically identical daughter cells.
- Occurs in somatic (body) cells.
- Involves one division cycle.

- Maintains the chromosome number (diploid).
- **Meiosis:**
 - Produces four genetically diverse daughter cells.
 - Occurs in germ (sex) cells.
 - Involves two division cycles.
 - Reduces the chromosome number by half (haploid).

7.Features of cell membrane and cell wall of plants:

- **Cell membrane:**
 - Phospholipid bilayer with embedded proteins.
 - Semi-permeable, controlling the movement of substances in and out of the cell.
- **Cell wall (plants):**
 - Composed mainly of cellulose.
 - Provides structural support and protection.
 - Maintains cell shape.

8. Cyclosis:

- Cyclosis, or cytoplasmic streaming, is the movement of the fluid substance (cytoplasm) within a plant or animal cell. This movement aids in the distribution of nutrients, organelles, and other materials within the cell.

9.Human epidermal cells:

- Human epidermal cells are the outermost layer of cells in the skin, providing a protective barrier against environmental factors.

10. Fungal cell:

- Fungal cells have a cell wall made of chitin, glucans, and proteins. They have organelles similar to those of animal cells, including a nucleus, mitochondria, and endoplasmic reticulum.

11. Define simple staining:

- Simple staining is a technique that uses a single dye to colour cells, making it easier to see cell shapes and arrangements under a microscope.

12. Why onion root is used for the study of mitosis?

- Onion root tips are used because they have a high rate of cell division, providing many cells in various stages of mitosis for observation.

13. Animal cell size, smallest and largest cell:

- Smallest cell: Mycoplasma (about 0.2-0.3 micrometers).
- Largest cell: The ostrich egg (can be up to 15 cm in diameter).

14. Fine adjustment and Coarse adjustment:

- Coarse adjustment: Used to bring the specimen into general focus.

- Fine adjustment: Used to fine-tune the focus and increase the detail of the specimen.

15. Function of Cell Wall:

- Provides structural support and protection.
- Maintains cell shape.
- Prevents excessive water intake.

16. Characteristics of Eukaryotic cell:

- Contains a nucleus.
- Has membrane-bound organelles.
- Larger than prokaryotic cells.
- Complex cytoskeleton.
- Undergoes mitosis and meiosis.

17. Name two main components of staining:

- The two main components are the dye (stain) and the mordant (which fixes the dye to the cell or tissue).

18. Write three subphases of interphase and also write their respective functions:

- G1 phase: Cell growth and preparation for DNA replication.
- S phase: DNA replication.
- G2 phase: Further growth and preparation for mitosis.

19. Difference between cell wall of plant and fungal cells:

- Plant cell wall: Composed mainly of cellulose.
- Fungal cell wall: Composed mainly of chitin.

20. Cytokinesis:

- The process by which the cytoplasm of a parent cell is divided between two daughter cells, usually occurring after mitosis or meiosis.

21. Which mordants are used in gram staining and why?

- Iodine is used as a mordant in Gram staining to form a complex with the crystal violet stain, which helps to fix the dye within the bacterial cell wall.

22. Function of SER and RER:

- Smooth Endoplasmic Reticulum (SER): Involved in lipid synthesis, detoxification, and calcium ion storage.
- Rough Endoplasmic Reticulum (RER): Involved in protein synthesis and folding, with ribosomes attached to its surface.

23. Significance of chromoplast:

- Chromoplasts are involved in the synthesis and storage of pigments, which give colour to fruits and flowers and help attract pollinators.

24. Principle of Gram staining:

- The principle of Gram staining is to differentiate bacteria based on the composition of their cell walls. Gram-positive

bacteria retain the crystal violet stain, appearing purple, while Gram-negative bacteria do not, appearing pink or red after counterstaining.

25. Gram staining types:

- Gram-positive staining.
- Gram-negative staining.

26. Define microscope:

- A microscope is an instrument used to view objects that are too small to be seen by the naked eye by magnifying them.

27. Explain types of Leucoplast:

- Leucoplasts are non-pigmented organelles in plant cells and include:

- Amyloplasts: Store starch.
- Elaioplasts: Store fats and oils.
- Proteinoplasts: Store proteins.

28. Acidic and Basic dye:

- Acidic dyes: Have a negative charge and bind to positively charged cell components.

- Basic dyes: Have a positive charge and bind to negatively charged cell components.

29. Compare and contrast Mitochondria and plastid:

- Mitochondria:
 - Powerhouse of the cell, producing ATP through cellular respiration.
 - Found in almost all eukaryotic cells.
 - Double-membraned.
- Plastid:
 - Involved in the synthesis and storage of food.
 - Found in plant cells and algae.
 - Includes chloroplasts, chromoplasts, and leucoplasts.
 - Double-membraned.

30.What are different regions of onion root tips?

- Root cap: Protects the growing tip.
- Meristematic zone: Region of active cell division.
- Elongation zone: Cells increase in size.
- Maturation zone: Cells differentiate into various types.

31.Components of Staining:

- The primary stain (e.g., crystal violet).
- The mordant (e.g., iodine).
- The decolorizer (e.g., alcohol).
- The counterstain (e.g., safranin).

32.Differential Staining:

- Differential staining uses more than one chemical stain to differentiate between different microorganisms or structures within a cell, such as Gram staining.

33.Explain Prophase of mitosis:

- In prophase, the chromatin condenses into visible chromosomes, the nuclear envelope breaks down, and the mitotic spindle begins to form.

34. Significance of cell division

- Cell division is crucial for growth, development, tissue repair, and reproduction in living organisms.

35.Explain morphology of cocci and Spirochetes:

- Cocci: Spherical-shaped bacteria, can be found singly, in pairs (diplococci), chains (streptococci), or clusters (staphylococci).

- Spirochetes: Spiral-shaped bacteria that are flexible and have a unique mode of movement.

36.Characteristics of prokaryotes:

- Lack a nucleus and membrane-bound organelles.
- Have a single, circular chromosome.
- Reproduced by binary fission.
- Smaller than eukaryotic cells.

- Have cell walls, often containing peptidoglycan.

37. Types of Staining:

- Simple staining.
- Differential staining (e.g., Gram staining).
- Special staining (e.g., spore staining, flagella staining).

38. Significance in flowers staining of chromoplast:

- Staining chromoplasts in flowers helps to study pigment distribution and understand their role in attracting pollinators.

39. Overview of Meiosis 2:

- Meiosis 2 resembles mitosis, where the sister chromatids of each chromosome are separated into different cells.

